What the data tells HealthConnect

Final analytics and recommendations

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The problem and my part

How can HealthConnect use data to reduce missed appointments and improve patient support?

Problem

Clinics lose money and patient goodwill, and need to see where.

My job

Turn the data into clear numbers, tested findings and actions.

Reality check

The data has no attendance field, so I worked on money, claims, staff and ratings.

Honest starting point: I can measure money, claims, workload and ratings. I cannot measure no-shows.

How I got here, Week 5 to Week 8

Each week built on the one before

Week 5
SwiftLog practice

Week 6
Database, dashboard, report

Week 7
120 tests, fixes

Week 8
Final KPIs and actions

Tools: PostgreSQL (SQL), Power BI (DAX), Python (checks), Excel

Data: 27 datasets, 20 hospitals, year 2026

Final KPIs

Recalculated again in Week 8: all 11 match

RWF 9.31 B	90.1%	RWF 920 M
total billed	collected	still unpaid
50%	**2.99 / 5**	**40.4%**
claims pending	average rating	poor ratings

Also: 20,000 patients, 5,000 detailed appointments, 11.7% admitted, 1,250 staff.

What I can prove, and what I cannot

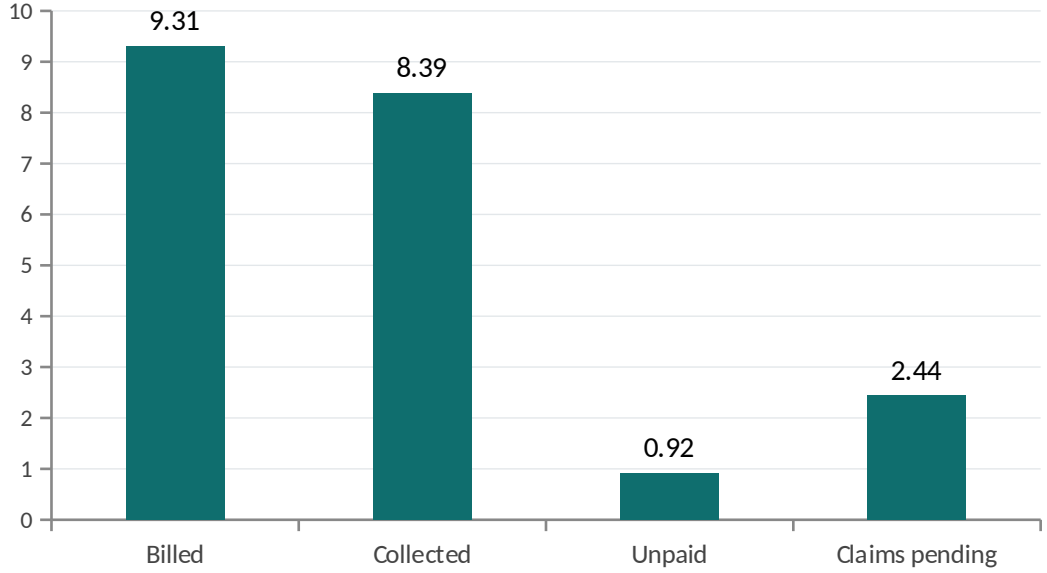
A finding counts only if a test says it is probably not chance

Finding	Result	Use it?
RWF 920 M unpaid	Counted from bills	**Yes**
Half of claims pending	Same for every insurer	**Yes**
Doctor workload differs	p = 0.010, small	**Yes, with care**
Busiest hospitals least liked	p = 0.91	**No**
Hospitals rated differently	p = 0.79	**No**
ICU worst, Kigali lowest, hypertension rising	p = 0.88, 0.32, 0.22	**No**

I removed the weak findings from Week 6 instead of defending them.

Money is the clearest finding

RWF billion, 2026



Unpaid: 11,887 bills not fully paid.

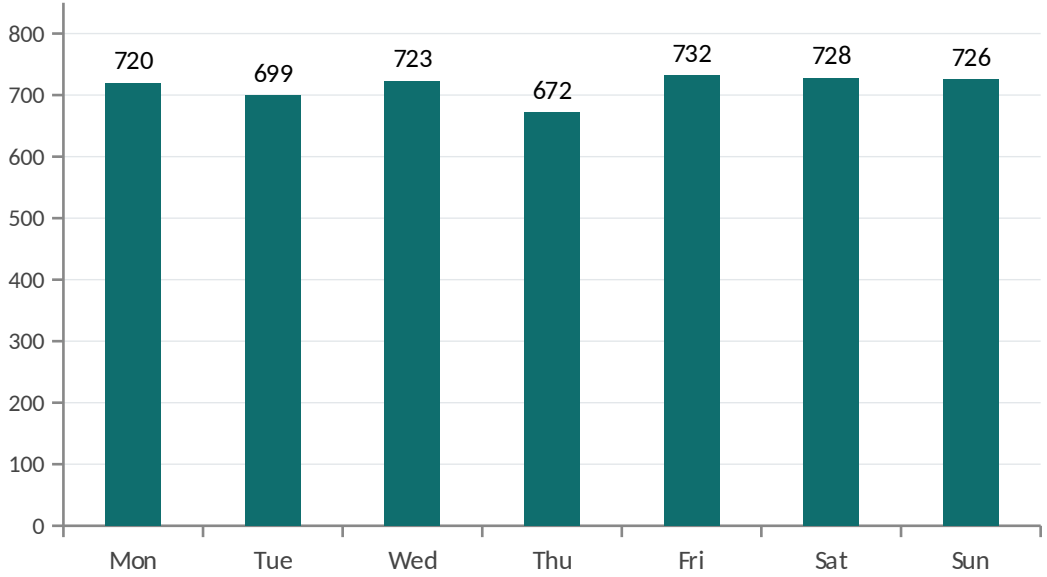
Claims: 21,028 of 42,036 pending, the same across insurers, payment methods and visit types.

So: the claims process needs a review, not one insurer.

Recovering unpaid bills and claims is cash the hospitals already earned.

Appointment patterns

No attendance field, so I report when appointments are booked



No busy day: weekends are as busy as weekdays (p = 0.69).

Flat all year: 400 to 440 a month.

Missing: whether patients came, cancelled or did not show.

First action: add an attendance status to every appointment.

Recommendations

Five actions, each tied to evidence

Action	Because
1. Record attendance for every appointment	No-shows cannot be measured today
2. Chase unpaid bills, largest first	RWF 920 M owed
3. Review the claims process end to end	Half pending, for every insurer
4. Staff evenly, including weekends	Demand is flat every day
5. Fix the feedback form, then re-measure	Stars and comments disagree in 6,336 records

Actions 2 and 3 recover money now. Action 1 makes the project goal measurable.

How it fits with the other tracks

What was actually exchanged

Track	Exchange	Status
Project Mgmt	Week 7: received model results, sent data facts, she acknowledged	Done
Data Science	My validated findings: claims have nothing to predict from; no attendance field	Sent. Use not yet confirmed
ML / GenAI	KPIs and limits available	Not yet

Result: effort on the claims model should go to a better target or fields, not tuning.

Limits and what I learned

Said plainly

Limits

- No attendance field; only 5,000 of 60,000 appointments link to a hospital
- Hospitals are codes only; ratings and comments disagree
- Dashboard file still to be saved from Power BI Desktop

What I learned

- Check numbers twice. Test before you claim. Say what the data cannot do.

Key takeaway: HealthConnect can recover money now, and must record attendance to fix missed appointments.